Fernando Contreras
CSD 380 Module 2.2 Assignment: Case Study: Operation InVersion at LinkedIn (2011)
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Repository: https://github.com/fernandojosecc/csd380-devops

Operation InVersion at LinkedIn (2011)

Operation InVersion is the case study in Chapter 6 of The DevOps Handbook. It shows what can happen when technical debt gets so bad that a company has to stop everything to fix it. The author uses it to make one main point: teams should pay down technical debt as part of their daily work, not wait for a crisis.

LinkedIn started in 2003 and ran mostly on Leo, a single large Java application that served every page and talked to Oracle databases. By 2010, most new work was happening in separate services, with nearly one hundred running outside of Leo. But Leo itself was only deployed every two weeks, and it kept going down in production. It was hard to troubleshoot, hard to recover, and hard to release new code. Adding more memory and CPUs did not fix it. One senior engineering manager, Josh Clemm, said it was clear they needed to "kill Leo" and break it into many small services. By fall 2011, six months after the company's IPO, releases were so painful that adding a bunch of new things at once could crash the site and keep engineers working late into the night.

Kevin Scott, decided to stop all work on new features for two months. The whole engineering department focused on fixing the core infrastructure, the deployments, and the architecture. They called it Operation InVersion. Scott saw it as a way to start building a new engineering culture, and he admitted it was scary to tell management there would be nothing new for two months right after going public.

LinkedIn built tools that let engineers create a new service, have automated systems check it for bugs and problems, and launch it straight to the live site. Instead of waiting weeks, they could upgrade the site three times a day. The number of services grew from about 150 in 2010 to over 750. There were fewer late-night emergencies, and more time to build new features.

Lessons Learned

1. Finding and fixing issues as they come up helps a company avoid these near-death experiences.
2. Breaking up the Leo monolith into many small services made faster and safer releases possible.
3. This chapter explains that if teams skip this, debt keeps growing until all their time goes to fixing fragile systems and feature delivery stops. InVersion was the extreme version of that. LinkedIn had to pay off nearly a decade of debt in one big push.
4. Once deployments were safe, the late nights went away and engineers could focus on new, useful features. Less pressure from technical debt also helps reduce burnout. A safer system of work helps people.
5. It takes leadership and real trade-offs. InVersion meant two months of work on non-functional requirements, at the expense of features promised to the public markets during the IPO. Scott's view was that an engineering team's job is to help the company win, and at that moment that meant fixing the foundation.

References

Kim, G., Humble, J., Debois, P., Willis, J., & Forsgren, N. (2021). *The DevOps handbook: How to create world-class agility, reliability, & security in technology organizations* (2nd ed.). IT Revolution Press.